

Trevor's problem

(Dated: October 5, 2025)

We have the average fitness function

$$\bar{W}(n, \bar{y}) = \sqrt{\frac{\gamma}{2\pi}} (a_f n + \gamma c w^2 \bar{y}) \exp\left(-\frac{\gamma \bar{y}^2}{2}\right) \quad (1)$$

The average date of breeding $\bar{y}$ is an affine function of $n > 0$: $\bar{y} = \bar{a} + \bar{b}n$. In order to estimate parameters $\bar{a}$ and $\bar{b}$, we can compute the average of $\bar{W}$ over the distribution of n . For that purpose, we need to make an assumption on the distribution of n . Then, once the average is obtained, we try to solve the equations for the critical points (in $\bar{a}$ and $\bar{b}$) in terms of the parameters characterizing the distributions.

Assuming that the distribution of n is exponential, $p(n) = \lambda e^{-\lambda n}$, then we can compute the average in terms of error functions. Given that $\bar{y} = \bar{a} + \bar{b}n$, we can write:

$$\bar{W}(\bar{a}, \bar{b}) = \int_0^\infty \bar{W}(n, \bar{a} + \bar{b}n) p(n) dn \quad (2)$$

Equivalently:

$$\bar{W}(\bar{a}, \bar{b}) = \lambda \sqrt{\frac{\gamma}{2\pi}} \int_0^\infty (rn + q) e^{-\frac{\gamma}{2}(\bar{a} + \bar{b}n)^2 - \lambda n} dn \quad (3)$$

with the definitions $r = a_f + \gamma c w^2 \bar{b}$ and $q = \gamma c w^2 \bar{a}$. Then,

$$\bar{W}(\bar{a}, \bar{b}) = \frac{e^{-\bar{a}^2 \gamma / 2}}{\bar{b}^3 \sqrt{2\gamma^3}} \left(r \bar{b} \sqrt{2\gamma} + \sqrt{\pi} [\bar{b}(\bar{b}q - \bar{a}r)\gamma - r\lambda] e^{(\bar{a}\bar{b}\gamma + \lambda)^2 / (2\gamma\bar{b}^2)} \operatorname{erfc}\left(\frac{\bar{a}\bar{b}\gamma + \lambda}{\bar{b}\sqrt{2\gamma}}\right) \right) \quad (4)$$

where we assumed that $\bar{b} > 0$. Here, erfc is the complementary error function, defined as $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$, for $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.

In this way, one can search for the critical points of $\bar{W}(\bar{a}, \bar{b})$ to identify parameters $\bar{a}$ and $\bar{b}$ in terms of γ , a_f , cw^2 and λ . When computing the gradient of $\bar{W}$, it is important to note that r is an affine function of $\bar{b}$ and q is a linear function of $\bar{a}$. Given the form of the function $\bar{W}(\bar{a}, \bar{b})$, a priori there is no reason why the equations for the critical points might lead to identifiability problems.

The system of equations we obtain for $\bar{a}$ and $\bar{b}$ ($\partial \bar{W} / \partial \bar{a} = 0$, $\partial \bar{W} / \partial \bar{b} = 0$) is non-linear, and has to be solved numerically.